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Data Transmission Method and Data Transmission Apparatus

Abstract

A generated data frame includes N symbols in a polarization direction. Every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, where $N=M \times Q$, Q is an even number, and M is an integer greater than or equal to 1. Q pilot symbols are generated by using the target polynomial and a seed. Each pilot symbol is one of $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, where A is a real number. The Q pilot symbols are direct current balanced. A degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of International Patent Application No. PCT/CN2023/116034 filed on Aug. 31, 2023, which claims priority to Chinese Patent Application No. 202211495134.6 filed on Nov. 26, 2022 and Chinese Patent Application No. 202310668432.9 filed on Jun. 6, 2023. All of the aforementioned patent applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates to the communication field, and in particular, to a data transmission method and a data transmission apparatus.

BACKGROUND

[0003] With continuous promotion of 5G, cloud computing, big data, and artificial intelligence, high-speed optical transport networks are developing toward high capacity, packetization, and intelligence. A coherent optical communication system uses amplitudes, phases, polarization, and frequencies of optical waves to carry information. To resist optical signal distortion caused by dispersion, polarization-dependent impairment, noise, non-linear effects, and other factors in a transmission process and maintain long-distance transmission, the coherent optical communication system usually adds some designed fixed symbol sequences to a transmission symbol sequence, so that a receiving end can restore a sent symbol.

[0004] An existing transmission symbol sequence is mainly used in a scenario of 400 gigabits per second (Gbps), and cannot adapt to future scenarios over 400 Gbps (including 600 Gbps, 800 Gbps, and the like). In addition, solutions in another technology have a problem that hardware implementation complexity is high, which needs to be resolved urgently in the future.

SUMMARY

[0005] Embodiments of this disclosure provide a data transmission method and a data transmission apparatus. A simple target polynomial is designed to generate pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the generated pilot symbols have good autocorrelation and cross-correlation characteristics, and are direct current balanced, which helps a receiving end restore signal quality.

[0006] According to a first aspect, an embodiment of this disclosure provides a data transmission method. The method applied to a transmitting end

includes the following steps: First, data frames are generated. In a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, $N=M \times Q$, Q is an even number, and M is an integer greater than or equal to 1. Q pilot symbols are generated by using a target polynomial and a seed, each pilot symbol is one of four complex numbers: $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, A is a real number, the Q pilot symbols are direct current balanced, a degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. Then, the data frames are sent. [0007] In the implementation, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N=M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0008] In some possible implementations, the target polynomial is one of the following: [0009] $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$; [0010] $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$; [0011] $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$; [0012] $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^6+x\{\text{circumflex over ()}\}^5+1$; [0013] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$; [0014] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^2+x+1$; [0015] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x+1$; [0016] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^2+1$; [0017] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^6+x+1$; [0018] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^7+x\{\text{circumflex over ()}\}^6+1$; [0019] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+1$; [0020] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^6+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^3+x+1$; [0021] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^7+x\{\text{circumflex over ()}\}^6+x\{\text{circumflex over ()}\}^2+x+1$; [0022] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^7+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^2+1$; [0023] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x\{\text{circumflex over ()}\}^2+1$; or [0024] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^7+x\{\text{circumflex over ()}\}^3+x\{\text{circumflex over ()}\}^2+1$.

[0025] In some possible implementations, the target polynomial is one of the following: [0026] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^7+x\{\text{circumflex over ()}\}^4+x+1$; or [0027] $x\{\text{circumflex over ()}\}^{10}+x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^6+x\{\text{circumflex over ()}\}^3+x\{\text{circumflex over ()}\}^2+x+1$.

[0028] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at the transmitting end and a laser at the receiving end.

[0029] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0030] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal to each other. This avoids a problem that the receiving end cannot distinguish between two polarization directions in actual transmission.

[0031] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following: [0032] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x115; [0033] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x02B; [0034] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x115; [0035] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x02B; [0036] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$, the seed in the first polarization direction: 0x115, and the seed in the second polarization direction: 0x08D; [0037] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^4+x\{\text{circumflex over ()}\}^3+x+1$, the seed in the first polarization direction: 0x08D, and the seed in the second polarization direction: 0x02B; [0038] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x0FE; [0039] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x0BF; [0040] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x17F; [0041] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x0FE; [0042] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x0BF; [0043] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x17F; [0044] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x0FE; [0045] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x0BF; [0046] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+x+1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x17F; [0047] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x11E; [0048] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x03D; [0049] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x08F; [0050] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x11E; [0051] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x03D; [0052] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x08F; [0053] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization direction: 0x11E, and the seed in the second polarization direction: 0x175; [0054] the target polynomial: $x\{\text{circumflex over ()}\}^9+x\{\text{circumflex over ()}\}^8+x\{\text{circumflex over ()}\}^5+x\{\text{circumflex over ()}\}^4+1$, the seed in the first polarization

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$-A+A_j, A+A_j, -A-A_j, A+A_j, A-A_j, -A+A_j, A-A_j, -A-A_j, A+A_j, A+A_j, -A-A_j, -A-A_j, \text{ and } A+A_j.$$

[0467] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[illegible][illegible]

[0472] In some possible implementations, a quantity of data frames is W, W is an integer greater than 1, and the W data frames are carried in a plurality of optical signals for transmission.

[0473] In some possible implementations, the W data frames are respectively carried in W optical signals. Any two of the W optical signals have different wavelengths. Alternatively, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[0474] In some possible implementations, the W data frames include a first data frame and a second data frame, and the first polarization direction and the second polarization direction are orthogonal to each other. In the first polarization direction, Q pilot symbols in the first data frame are generated by using a first target polynomial and a first seed; and in the second polarization direction, the Q pilot symbols in the first data frame are generated by using the first target polynomial and a second seed. In the first polarization direction, Q pilot symbols in the second data frame are generated by using a second target polynomial and a third seed; and in the second polarization direction, the Q pilot symbols in the second data frame are generated by using the second target polynomial and a fourth seed.

[0475] In some possible implementations, a first pilot symbol sequence in the first data frame is the same as a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0476] In some possible implementations, the first target polynomial is the same as the second target polynomial, the first seed is the same as the third seed, the second seed is the same as the fourth seed, the first seed is different from the second seed, and the third seed is different from the fourth seed.

[0477] In some possible implementations, a first pilot symbol sequence in the first data frame is different from a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0478] In some possible implementations, the first target polynomial is the same as the second target polynomial, and any two of the first seed, the second seed, the third seed, and the fourth seed are different from each other.

[0479] In some possible implementations, the first target polynomial is different from the second target polynomial.

[0480] In some possible implementations, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

[0481] In some possible implementations, in a polarization direction, the modulation format of the symbol in the data frame is QPSK, and $A=-1$ or 1 .

[0482] According to a fifth aspect, an embodiment of this disclosure provides a data transmission apparatus. The data transmission apparatus includes a processing unit and a sending unit. The processing unit is configured to generate data frames. In a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, $N=M \times Q$, Q is an even number, M is an integer greater than or equal to 1, Q pilot symbols are generated by using a target polynomial and a seed, each pilot symbol is one of four complex numbers: $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, A is a real number, the Q pilot symbols are direct current balanced, a degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. The sending unit is configured to send the data frames.

[0483] In the implementation, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N=M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0484] In some possible implementations, the target polynomial is one of the following: [0485] $x^9 + x^4 + x^3 + x + 1$; [0486] $x^9 + x^5 + x^4 + x + 1$; [0487] $x^9 + x^8 + x^5 + x^4 + 1$; [0488] $x^9 + x^8 + x^6 + x^5 + 1$; [0489] $x^{10} + x^4 + x^3 + x + 1$; [0490] $x^{10} + x^5 + x^2 + x + 1$; [0491] $x^{10} + x^8 + x^5 + x + 1$; [0492] $x^{10} + x^8 + x^2 + 1$; [0493] $x^{10} + x^9 + x^6 + x + 1$; [0494] $x^{10} + x^9 + x^7 + x^6 + 1$; [0495] $x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$; [0496] $x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$; [0497] $x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$; [0498] $x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$; [0499] $x^{10} + x^9 + x^8 + x^5 + x^4 + x^3 + x^2 + 1$; or [0500] In some possible implementations, the target polynomial is one of the following:

[0501] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at a transmitting end and a laser at the receiving end.

[0502] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0503] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal

[illegible]

[illegible]

[illegible]

different wavelengths. Alternatively, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[0710] In some possible implementations, the W data frames include a first data frame and a second data frame, and the first polarization direction and the second polarization direction are orthogonal to each other. In the first polarization direction, Q pilot symbols in the first data frame are generated by using a first target polynomial and a first seed; and in the second polarization direction, the Q pilot symbols in the first data frame are generated by using the first target polynomial and a second seed. In the first polarization direction, Q pilot symbols in the second data frame are generated by using a second target polynomial and a third seed; and in the second polarization direction, the Q pilot symbols in the second data frame are generated by using the second target polynomial and a fourth seed.

[0711] In some possible implementations, a first pilot symbol sequence in the first data frame is the same as a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0712] In some possible implementations, the first target polynomial is the same as the second target polynomial, the first seed is the same as the third seed, the second seed is the same as the fourth seed.

[0713] In some possible implementations, a first pilot symbol sequence in the first data frame is different from a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0714] In some possible implementations, the first target polynomial is the same as the second target polynomial, and any two of the first seed, the second seed, the third seed, and the fourth seed are different from each other.

[0715] In some possible implementations, the first target polynomial is different from the second target polynomial.

[0716] In some possible implementations, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

[0717] In some possible implementations, in a polarization direction, a modulation format of the symbol in the data frame is 16QAM, and $A = -1, 1, -3, 3, -\sqrt{5}, \sqrt{5}$.

[0718] In some possible implementations, in a polarization direction, the modulation format of the symbol in the data frame is QPSK, and $A = -1$ or 1 .

[0719] According to a seventh aspect, an embodiment of this disclosure provides a data transmission apparatus. The data transmission apparatus includes a receiving unit and a processing unit. The receiving unit is configured to receive data frames. In a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, $N = M \times Q$, Q is an even number, M is an integer greater than or equal to 1, Q pilot symbols are generated by using a target polynomial and a seed, each pilot symbol is one of four complex numbers: $-A - Aj$, $-A + Aj$, $A - Aj$, and $A + Aj$, A is a real number, the Q pilot symbols are direct current balanced, a degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. The processing unit is configured to process the data frames.

[0720] In the implementation, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N = M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0721] In some possible implementations, the target polynomial is one of the following: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$; [0722] $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^4 + x + 1$; [0723] $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^4 + 1$; [0724] $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^6 + x \{ \text{circumflex over ()} \}^5 + 1$; [0725] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$; [0726] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^2 + x + 1$; [0727] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^5 + x + 1$; [0728] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x + 1$; [0729] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^2 + 1$; [0730] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^6 + x + 1$; [0731] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^7 + x \{ \text{circumflex over ()} \}^6 + 1$; [0732] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^5 + 1$; [0733] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^6 + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^3 + x + 1$; [0734] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^7 + x \{ \text{circumflex over ()} \}^6 + x \{ \text{circumflex over ()} \}^2 + x + 1$; [0735] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^7 + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^2 + 1$; [0736] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x \{ \text{circumflex over ()} \}^2 + 1$; or [0737] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^7 + x \{ \text{circumflex over ()} \}^3 + x \{ \text{circumflex over ()} \}^2 + 1$.

[0738] In some possible implementations, the target polynomial is one of the following: [0739] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^8 + x \{ \text{circumflex over ()} \}^7 + x \{ \text{circumflex over ()} \}^4 + x + 1$; or [0740] $x \{ \text{circumflex over ()} \}^{10} + x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^6 + x \{ \text{circumflex over ()} \}^3 + x \{ \text{circumflex over ()} \}^2 + x + 1$.

[0741] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at a transmitting end and a laser at the receiving end.

[0742] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0743] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal to each other. This avoids a problem that the receiving end cannot distinguish between two polarization directions in actual transmission.

[0744] In some possible implementations, $N = 6144$, $M = 64$, $Q = 96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following: [0745] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x115; [0746] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x02B; [0747] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x115; [0748] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x02B; [0749] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$, the seed in the first polarization direction: 0x115, and the seed in the second polarization direction: 0x08D; [0750] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^4 + x \{ \text{circumflex over ()} \}^3 + x + 1$, the seed in the first polarization direction: 0x08D, and the seed in the second polarization direction: 0x02B; [0751] the target polynomial: $x \{ \text{circumflex over ()} \}^9 + x \{ \text{circumflex over ()} \}^5 + x \{ \text{circumflex over ()} \}^4 + x + 1$, the seed in the first polarization direction: 0x098, and the seed

[illegible]

[illegible]

[illegible]

the first polarization direction and the second polarization direction are orthogonal to each other.

[0908] 96 pilot symbols in the first polarization direction are sequentially: [0909] A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, A-Aj, -A+Aj, -A+Aj, A-Aj, -A+Aj, A+Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj, A+Aj, -A-Aj, A+Aj, A+Aj, -A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, -A-Aj, A+Aj, A+Aj, A+Aj, A+Aj, A+Aj, A-Aj, -A-Aj, -A-Aj, A-Aj, -A+Aj, -A+Aj, A-Aj, A-Aj, A-Aj, A+Aj, A+Aj, A+Aj, A+Aj, A+Aj, A-Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, -A+Aj, A-Aj, -A+Aj, -A+Aj, A+Aj, A+Aj, A-Aj, A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, -A-Aj, and A+Aj.

[illegible]

[0913] In some possible implementations, $N=6144$, $M=64$, $Q=96$, the target polynomial is $x^{\circlearrowleft}10+x^{\circlearrowleft}9+x^{\circlearrowleft}4+x+1$, the seed in the first polarization direction is 0x12A, the seed in the second polarization direction is 0x039, and the first polarization direction and the second polarization direction are orthogonal to each other.

[illegible][illegible]

[0918] In some possible implementations, a quantity of data frames is W, W is an integer greater than 1, and the W data frames are carried in a plurality of optical signals for transmission.

[0919] In some possible implementations, the W data frames are respectively carried in W optical signals. Any two of the W optical signals have different wavelengths. Alternatively, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[0920] In some possible implementations, the W data frames include a first data frame and a second data frame, and the first polarization direction and the second polarization direction are orthogonal to each other. In the first polarization direction, Q pilot symbols in the first data frame are generated by using a first target polynomial and a first seed; and in the second polarization direction, the Q pilot symbols in the first data frame are generated by using the first target polynomial and a second seed. In the first polarization direction, Q pilot symbols in the second data frame are generated by using a second target polynomial and a third seed; and in the second polarization direction, the Q pilot symbols in the second data frame are generated by using the second target polynomial and a fourth seed.

[0921] In some possible implementations, a first pilot symbol sequence in the first data frame is the same as a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0922] In some possible implementations, the first target polynomial is the same as the second target polynomial, the first seed is the same as the third seed, the second seed is the same as the fourth seed, the first seed is different from the second seed, and the third seed is different from the fourth seed.

[0923] In some possible implementations, a first pilot symbol sequence in the first data frame is different from a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0924] In some possible implementations, the first target polynomial is the same as the second target polynomial, and any two of the first seed, the second seed, the third seed, and the fourth seed are different from each other.

[0925] In some possible implementations, the first target polynomial is different from the second target polynomial.

[0926] In some possible implementations, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

[0927] In some possible implementations, in a polarization direction, the modulation format of the symbol in the data frame is QPSK, and $A=-1$ or 1 .

[0928] According to an eighth aspect, an embodiment of this disclosure provides a data transmission apparatus. The data transmission apparatus includes a receiving unit and a processing unit. The receiving unit is configured to receive data frames. In a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, $N=M \times Q$, Q is an even number, M is an integer greater than or equal to 1, Q pilot symbols are generated by using a target polynomial and a seed, each pilot symbol is one of four complex numbers: $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, A is a real number, the Q pilot symbols are direct current balanced, and a difference between every two of quantities of pilot symbols that are respectively $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$ in the data frame is less than or equal to 2. The processing unit is configured to process the data frames.

[0929] In the implementation, the difference between the every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in the data frame is less than or equal to 2. In addition, the quantity of pilot symbols that are $-A-A_j$ is the same as the quantity of pilot symbols that are $A+A_j$, and the quantity of pilot symbols that are $-A+A_j$ is the same as the quantity of pilot symbols that are $A-A_j$. This effectively ensures that symbol quantities in each polarization direction are approximately balanced, and can further ensure that a sequence including the pilot symbols achieves the direct current balance, to help a receiving end restore signal quality.

[0930] In some possible implementations, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor + 1$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor + 1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$. $\lfloor a \rfloor$ represents rounding down a positive real number a to the nearest integer.

second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0955] In some possible implementations, the first target polynomial is the same as the second target polynomial, and any two of the first seed, the second seed, the third seed, and the fourth seed are different from each other.

[0956] In some possible implementations, the first target polynomial is different from the second target polynomial.

[0957] In some possible implementations, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

[0958] In some possible implementations, in a polarization direction, the modulation format of the symbol in the data frame is QPSK, and $A = -1$ or 1 .

[0959] In embodiments of this disclosure, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N = M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that the simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, the simple hardware structure may be used for the implementation. In addition, the pilot symbols generated in the foregoing manner have the good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps the receiving end restore the signal quality.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0960] FIG. 1 is a diagram of a communication system to which an embodiment of this disclosure is applied;

[0961] FIG. 2 is a diagram of an implementation of a transmitting-end DSP processor according to an embodiment of this disclosure;

[0962] FIG. 3 is a diagram of another implementation of a transmitting-end DSP processor according to an embodiment of this disclosure;

[0963] FIG. 4 is a schematic flowchart of a data transmission method according to an embodiment of this disclosure;

[0964] FIG. 5 is a diagram of a structure of a data frame according to an embodiment of this disclosure;

[0965] FIG. 6 is a diagram of symbols in a constellation diagram according to an embodiment of this disclosure;

[0966] FIG. 7 is a diagram of a first-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0967] FIG. 8 is a diagram of a second-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0968] FIG. 9 is a diagram of a structure of another data frame according to an embodiment of this disclosure;

[0969] FIG. 10 is a diagram of a third-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0970] FIG. 11 is a diagram of a first implementation procedure of generating pilot symbols based on dual-polarization (DP)-16QAM modulation according to an embodiment of this disclosure;

[0971] FIG. 12A to FIG. 12C are diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure;

[0972] FIG. 13A and FIG. 13B are diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0973] FIG. 14 is a diagram of a fourth-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0974] FIG. 15 is a diagram of a second implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0975] FIG. 16A to FIG. 16C are other diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure;

[0976] FIG. 17A and FIG. 17B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0977] FIG. 18 is a diagram of a transmission scenario of a plurality of data frame streams according to an embodiment of this disclosure;

[0978] FIG. 19 is a diagram of a fifth-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0979] FIG. 20 is a diagram of a third implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0980] FIG. 21A to FIG. 21C are other diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure;

[0981] FIG. 22A and FIG. 22B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0982] FIG. 23 is a diagram of a sixth-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0983] FIG. 24 is a diagram of a fourth implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0984] FIG. 25A and FIG. 25B are other diagrams of an autocorrelation characteristic and a cross-correlation characteristic that correspond to pilot symbols according to an embodiment of this disclosure;

[0985] FIG. 26A and FIG. 26B are diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0986] FIG. 27 is a diagram of a seventh-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0987] FIG. 28 is a diagram of a fifth implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0988] FIG. 29A and FIG. 29B are other diagrams of a correlation characteristic and a cross-correlation characteristic that correspond to pilot symbols according to an embodiment of this disclosure;

[0989] FIG. 30A and FIG. 30B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0990] FIG. 31 is a diagram of a structure of a data transmission apparatus used at a transmitting end according to an embodiment of this disclosure;

[0991] FIG. 32 is a diagram of a structure of a data transmission apparatus used at a receiving end according to an embodiment of this disclosure; and

[0992] FIG. 33 is a diagram of another structure of a data transmission apparatus according to an embodiment of this disclosure.

DESCRIPTION OF EMBODIMENTS

[0993] Embodiments of this disclosure provide a data transmission method and a data transmission apparatus. A simple target polynomial is designed to generate pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the generated pilot symbols have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0994] It should be noted that in the specification, claims, and the foregoing accompanying drawings of this disclosure, the terms “first”, “second”, and the like are intended to distinguish between similar objects but do not limit a specific order or sequence. It should be understood that the foregoing terms may be interchanged in proper cases, so that embodiments described in this disclosure can be implemented in an order other than the content described in this disclosure. In addition, the terms “include”, “have”, and any other variant thereof are intended to cover a non-exclusive

inclusion. For example, a process, a method, a system, a product, or a device that includes a series of steps or units is not necessarily limited to those expressly listed steps or units, but may include other steps or units not expressly listed or inherent to the process, the method, the product, or the device.

[0995] FIG. 1 is a diagram of a communication system to which an embodiment of this disclosure is applied. As shown in FIG. 1, at a transmitting end, a source provides a to-be-sent data stream. An encoder receives and encodes the data stream. Encoded codeword information combining a parity bit and an information bit is sent to a transmitting-end digital signal processing (DSP) processor for framing, and is transmitted through a channel to a receiving end. After receiving a signal with distortion resulting from noise or other impairments on the channel, the receiving end sends the signal to a receiving-end DSP processor for dispersion compensation, synchronization, phase recovery, and other operations. Then, a decoder performs decoding to restore original data, and sends the data to a sink.

[0996] FIG. 2 is a diagram of an implementation of a transmitting-end DSP processor according to an embodiment of this disclosure. As shown in FIG. 2, in a possible implementation, the transmitting-end DSP processor performs symbol mapping on a received data sequence. Usually, the received data sequence is an information sequence and a parity sequence obtained based on a forward error correction (FEC) code. A symbol mapping manner includes but is not limited to QPSK and QAM. The transmitting-end DSP processor further performs polarization symbol distribution on the data sequence to obtain a DP symbol, for example, a DP-QPSK, DP-8QAM, DP-16QAM, DP-32QAM, or DP-64QAM symbol. For ease of description, two polarization directions are collectively denoted as an X polarization direction and a Y polarization direction below, where the X polarization direction and the Y polarization direction are orthogonal to each other. It should be understood that the X polarization direction and the Y polarization direction are not two specified polarization directions, but are any two polarization directions that are orthogonal to each other.

[0997] Further, the transmitting-end DSP processor performs the following framing processing on a specific quantity of DP symbols. Pilot symbols are inserted in each of the X polarization direction and the Y polarization direction, to obtain a to-be-sent DP symbol sequence, which is referred to as a frame. In embodiments, the frame is referred to as a data frame, and may also be referred to as a DSP frame.

[0998] It should be noted that, after the symbol mapping, the symbols may be further interleaved, and the foregoing framing processing is performed on interleaved symbols. It should be understood that one DP symbol may be represented by two symbols, where one symbol is located in the X polarization direction, and the other symbol is located in the Y polarization direction. Each symbol may be represented by a complex number. For example, a symbol obtained through 16QAM modulation may be represented by any one of the following 16 complex numbers: $\pm 1 \pm j$, $\pm 1 \pm 3j$, $\pm 3 \pm j$, and $\pm 3 \pm 3j$, where $\pm$ herein represents a positive value or a negative value. For example, $+3$ represents 3 or -3 . j represents an imaginary unit. In some scenarios, the imaginary unit may alternatively be represented by another symbol like i . This is not limited herein. In some cases, real and imaginary parts are normalized, but the essence does not change. Further, a sequence having L DP symbols may be completely represented by two complex sequences whose lengths are L , where one complex sequence represents a symbol in the X polarization direction, and the other complex sequence represents a symbol in the Y polarization direction. Each complex sequence whose length is L is represented by a real part sequence (also referred to as an I-channel sequence) whose length is L and an imaginary part sequence (also referred to as a Q-channel sequence) whose length is L , where L is an integer greater than 1. Therefore, there are four different types of sequences, including an X-polarization I-channel sequence, an X-polarization Q-channel sequence, a Y-polarization I-channel sequence, and a Y-polarization Q-channel sequence.

[0999] FIG. 3 is a diagram of another implementation of a transmitting-end DSP processor according to an embodiment of this disclosure. As shown in FIG. 3, different from the framing operation performed on the symbols shown in FIG. 2, in another possible implementation, before symbol mapping, the transmitting-end DSP processor inserts, in a received data sequence according to a used symbol mapping rule, bits corresponding to pilot symbols, and then may obtain, through the symbol mapping and polarization symbol distribution, a same frame as that in the operation shown in FIG. 2. In this case, before the symbol mapping, a bit sequence obtained by inserting the bits corresponding to the symbols may be further interleaved, and then the same frame as that in the operation shown in FIG. 2 is obtained through the symbol mapping and the polarization symbol distribution.

[1000] It should be noted that a specific framing manner used by the transmitting-end DSP processor is not limited in this disclosure. In addition to the framing manners described in FIG. 2 and FIG. 3, other similar framing manners are also applicable to this solution, and are not described one by one herein.

[1001] FIG. 4 is a schematic flowchart of a data transmission method according to an embodiment of this disclosure. It should be understood that the data transmission method is applied to a transmitting end, for example, may be implemented by the transmitting-end DSP processor shown in FIG. 1.

[1002] 401: Generate data frames.

[1003] It should be noted that a specific implementation of generating the data frames is not limited in this disclosure. For example, the framing manner described in FIG. 2 or FIG. 3 may be used. Certainly, other similar framing manners are also applicable to this solution, and are not described one by one herein. It should be understood that the data frame includes symbols in two polarization directions, and structures of the data frame in the two polarization directions are similar. For example, the data frame includes N symbols in an X polarization direction and N symbols in a Y polarization direction. The following describes the structure of the data frame by using one of the polarization directions as an example.

[1004] FIG. 5 is a diagram of a structure of a data frame according to an embodiment of this disclosure. As shown in FIG. 5, in a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and $M-1$ payload symbols, $N=M \times Q$, Q is an even number, and M is an integer greater than or equal to 1. In other words, the every M consecutive symbols in the N symbols may be considered as one group, and the N symbols include Q groups of symbols in total. For example, $N=6144$, $M=64$, $Q=96$, a symbol 1 to a symbol 64 are 64 consecutive symbols in a 1st group, a symbol 65 to a symbol 128 are 64 consecutive symbols in a 2nd group, . . . , and a symbol 6081 to a symbol 6144 are 64 consecutive symbols in a 96th group. It should be understood that the payload symbol may also be referred to as a symbol before framing, and includes an information symbol and a parity symbol that are obtained through FEC encoding. At a receiving end, a pilot symbol may be used to assist in carrier phase recovery, and may be further used to distinguish between the two polarization directions.

[1005] It should be understood that a specific location of a pilot symbol in each group of M symbols is not limited in this disclosure. In an example, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located. For example, a 1st symbol in the data frame shown in FIG. 5 is a 1st pilot symbol.

[1006] It should be noted that Q pilot symbols in the data frame are generated by using a target polynomial and a seed. Each pilot symbol is one of $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, and A is a real number. A degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8, so that hardware implementation complexity is low. In addition, the Q pilot symbols are direct current balanced, that is, a sum of the Q pilot symbols is 0, to be specific, a sum of real parts and a sum of imaginary parts of complex numbers corresponding to the Q pilot symbols are both 0, so that the direct current balance can be achieved. This helps the receiving end restore signal quality.

[1007] 402: Send the data frames.

[1008] The sent data frames are transmitted through a channel to the receiving end. A specific operation performed after the receiving end receives the data frames is not described in detail in this disclosure. For details, refer to the diagram of the structure of the system shown in FIG. 1.

[1009] The following describes in detail a characteristic of the pilot symbol and a specific manner of generating the pilot symbol.

[1010] In embodiments of this disclosure, a value of A is determined based on a modulation format used during symbol generation. In some actual application scenarios, $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$ are symbols in a constellation diagram of the used modulation format. For example, if QPSK

is used, there are only four symbols. In this case, $A=+1$, and each pilot symbol may be represented by one of $-1j$, $-1+j$, $1-j$, and $1+j$. In one frame, all frame symbols represented by the four complex numbers exist. If 16QAM is used, there are 16 symbols in the constellation diagram. In this case, $A=+1$ or $+3$.

[1011] FIG. 6 is a diagram of symbols in a constellation diagram according to an embodiment of this disclosure. As shown in FIG. 6, when $A=3$ or -3 , each pilot symbol may be represented by one of $-3-3j$, $-3+3j$, $3-3j$, and $3+3j$, for example, a hollow symbol shown in FIG. 6. Similarly, if 64QAM is used, $A=+1$, $+3$, $+5$, or $+7$. It should be noted that a higher-order modulation format may alternatively be used. Details are not described in this disclosure. In an actual transmission process, four outer symbols in the constellation diagram are used as pilot symbols, so that a symbol error probability is low.

[1012] It should be noted that the symbols in the constellation diagram may alternatively be compressed, and correspondingly, a value of A is also compressed. The 16QAM is used as an example, and power normalization is performed on the 16 symbols in the 16QAM constellation diagram. In this case, values of the 16 symbols in the 16QAM constellation diagram are changed to

$$\{ \pm \frac{1}{\sqrt{10}} \pm \frac{1}{\sqrt{10}}j, \pm \frac{1}{\sqrt{10}} \pm \frac{3}{\sqrt{10}}j, \pm \frac{3}{\sqrt{10}} \pm \frac{1}{\sqrt{10}}j, \pm \frac{3}{\sqrt{10}} \pm \frac{3}{\sqrt{10}}j \},$$

and the value of A is

$$\{ 00002 \} \frac{1}{\sqrt{10}} \text{ or } \frac{3}{\sqrt{10}}.$$

[1013] Alternatively, normalization in another manner may be used. This is not limited in this disclosure.

[1014] It should be understood that, when four outermost symbols in the constellation diagram are used as the pilot symbols $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, the pilot symbols have high sensitivity, but have a large peak to average power ratio. When four innermost symbols in the constellation diagram are used as the pilot symbols $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, the pilot symbols have small noise, but have low sensitivity.

[1015] It should be noted that, in some actual application scenarios, alternatively, the pilot symbols $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$ may not be the symbols in the constellation diagram of the used modulation format, and may be four symbols in an intermediate area between the four outermost symbols and the four innermost symbols in the constellation diagram. In this case, the pilot symbols have fair noise and sensitivity, but have a low peak to average power ratio. The 16QAM is used as an example, the values of the 16 symbols in the 16QAM constellation diagram are $\{\pm 1\pm 1j, \pm 1\pm 3j, \pm 3\pm 1j, \pm 3\pm 3j\}$, and the value of the real number A satisfies $1 \leq A \leq 3$. For example, the real number $A=\sqrt{5}$.

[1016] In addition, the two polarization directions are orthogonal to each other. To be specific, when one of the polarization directions is X polarization, the other polarization direction is Y polarization; or when one of the polarization directions is Y polarization, the other polarization direction is X polarization.

[1017] In the data frame, a sequence including Q pilot symbols in the X polarization direction is different from a sequence including Q pilot symbols in the Y polarization direction. For example, if a sequence including pilot symbols in the X polarization direction is $-A-Aj$, $-A-Aj$, $A+Aj$, and $A-Aj$, in a same order, a sequence including pilot symbols in the Y polarization direction cannot be the same as the sequence including the pilot symbols in the X polarization direction, and may be $-A-Aj$, $-A-Aj$, $A+Aj$, and $A+Aj$. There is one different symbol. This avoids a problem that the receiving end cannot distinguish between the two polarization directions in actual transmission.

[1018] In embodiments of this disclosure, when a 10-degree polynomial is used as the target polynomial, the 10-degree polynomial may be represented as: [1019]

$$x_{\text{sup.10}}+a_{\text{sub.9}}x_{\text{sup.9}}+a_{\text{sub.8}}x_{\text{sup.8}}+a_{\text{sub.7}}x_{\text{sup.7}}+a_{\text{sub.6}}x_{\text{sup.6}}+a_{\text{sub.5}}x_{\text{sup.5}}+a_{\text{sub.4}}x_{\text{sup.4}}+a_{\text{sub.3}}x_{\text{sup.3}}+a_{\text{sub.2}}x_{\text{sup.2}}+a_{\text{sub.1}}x_{\text{sup.1}}+x_{\text{sup.0}}$$

where values of $a_{\text{sub.9}}$, $\dots$, and $a_{\text{sub.1}}$ may be 0 or 1, and a quantity of non zeros in $a_{\text{sub.9}}$, $\dots$, and $a_{\text{sub.1}}$ is not greater than 6. FIG. 7 is a diagram of a first-generation structure of a pilot symbol according to an embodiment of this disclosure. As shown in FIG. 7, each block may be considered as one storage element, and a quantity of storage elements is the same as a quantity of bits in a preloaded seed, that is, each storage element is configured to input a corresponding bit that is in the seed. For example, if a seed length is 10 bits, and the seed may be represented as $m_{\text{sub.9}}$, $m_{\text{sub.8}}$, $m_{\text{sub.7}}$, $m_{\text{sub.6}}$, $m_{\text{sub.5}}$, $m_{\text{sub.4}}$, $m_{\text{sub.3}}$, $m_{\text{sub.2}}$, $m_{\text{sub.1}}$, and $m_{\text{sub.0}}$ in a binary form, 10 corresponding storage elements are used. Certainly, the seed may alternatively be represented in a hexadecimal or decimal form, and needs to be converted into the binary form during an operation with the target polynomial. For example, 0110111000 is represented as 0x1B8 in the hexadecimal form, and is represented as 440 in the decimal form.

[1020] It should be noted that the polynomial $x_{\text{sup.10}}+a_{\text{sub.9}}x_{\text{sup.9}}+ag \ X$

$$x_{\text{sup.8}}+a_{\text{sub.7}}x_{\text{sup.7}}+a_{\text{sub.6}}x_{\text{sup.6}}+a_{\text{sub.5}}x_{\text{sup.5}}+a_{\text{sub.4}}x_{\text{sup.4}}+a_{\text{sub.3}}x_{\text{sup.3}}+a_{\text{sub.2}}x_{\text{sup.2}}+a_{\text{sub.1}}x_{\text{sup.1}}+1$$
 is sometimes written as $x_{\text{sup.10}}+a_{\text{sub.9}}(x_{\text{sup.9}})+a_{\text{sub.8}}(x_{\text{sup.8}})+a_{\text{sub.7}}(x_{\text{sup.7}})+a_{\text{sub.6}}(x_{\text{sup.6}})+a_{\text{sub.5}}(x_{\text{sup.5}})+a_{\text{sub.4}}(x_{\text{sup.4}})+a_{\text{sub.3}}(x_{\text{sup.3}})+a_{\text{sub.2}}(x_{\text{sup.2}})+a_{\text{sub.1}}(x_{\text{sup.1}})+1$

[1021] When a 9-degree polynomial is used as the target polynomial, the 9-degree polynomial may be represented as: [1022]

$$x_{\text{sup.9}}+a_{\text{sub.8}}x_{\text{sup.8}}+a_{\text{sub.7}}x_{\text{sup.7}}+a_{\text{sub.6}}x_{\text{sup.6}}+a_{\text{sub.5}}x_{\text{sup.5}}+a_{\text{sub.4}}x_{\text{sup.4}}+a_{\text{sub.3}}x_{\text{sup.3}}+a_{\text{sub.2}}x_{\text{sup.2}}+a_{\text{sub.1}}x_{\text{sup.1}}+1,$$
 where values of $a_{\text{sub.8}}$, $\dots$, and $a_{\text{sub.1}}$ may be 0 or 1, and a quantity of non zeros in $a_{\text{sub.8}}$, $\dots$, and $a_{\text{sub.1}}$ is not greater than 6.

[1023] FIG. 8 is a diagram of a second-generation structure of a pilot symbol according to an embodiment of this disclosure. As shown in FIG. 8, a seed length is 9 bits, and the seed may be represented as $m_{\text{sub.8}}$, $m_{\text{sub.7}}$, $m_{\text{sub.6}}$, $m_{\text{sub.5}}$, $m_{\text{sub.4}}$, $m_{\text{sub.3}}$, $m_{\text{sub.2}}$, $m_{\text{sub.1}}$, and $m_{\text{sub.0}}$ in a binary form. Certainly, the seed may alternatively be represented in a hexadecimal or decimal form, and needs to be converted into the binary form during an operation with the target polynomial.

[1024] It should be noted that an algebraic expression formed through addition (or subtraction) of several monomials is referred to as a polynomial. Each monomial in the polynomial is referred to as a term of the polynomial, and a highest degree (degree, which is also referred to as an order) in degrees of these monomials is a degree of the polynomial. A quantity of terms of the polynomial is a quantity of the foregoing monomials whose coefficients are not 0. For example, a quantity of terms of the foregoing 10-degree polynomial is equal to 2 plus the quantity of non zeros in $a_{\text{sub.9}}$, $\dots$, and $a_{\text{sub.1}}$.

[1025] In embodiments of this disclosure, a same target generator polynomial may be used for pilot symbols in two orthogonal polarization directions. However, because seeds used in the two polarization directions are different, correspondingly, pilot symbols output in the two polarization directions are not completely the same.

[1026] In FIG. 7 or FIG. 8, for a scenario in which Q pilot symbols need to be generated, a consecutive bit sequence $b_{\text{sub.0}}$, $b_{\text{sub.1}}$, $b_{\text{sub.2}}$, $\dots$, and $b_{\text{sub.2Q-1}}$ whose bit length is $2Q$ is obtained based on the target polynomial and the seed. The foregoing bit sequence is also referred to as a pseudo random binary sequence (PRBS). A bit sequence generated by using the 9-degree polynomial is also referred to as a PRBS9, and a bit sequence generated by using the 10-degree polynomial is also referred to as a PRBS10. Every two consecutive bits in the bit sequence $b_{\text{sub.0}}$, $b_{\text{sub.1}}$, $b_{\text{sub.2}}$, $\dots$, and $b_{\text{sub.2Q-1}}$ are mapped to one symbol, where $b_{\text{sub.2t}}$ and $b_{\text{sub.2t+1}}$ are mapped to one symbol $(2b_{\text{sub.2t-1}})A+(2b_{\text{sub.2t}}-1)Aj$, and $0 \leq t < Q$.

[1027] In this embodiment of this disclosure, the target polynomial and the seed may be determined by designing the values of the coefficients $a_{\text{sub.9}}$, $\dots$, and $a_{\text{sub.1}}$ in the 10-degree polynomial or the values of the coefficients $a_{\text{sub.8}}$, $\dots$, and $a_{\text{sub.1}}$ in the 9-degree polynomial, so that a generated pilot symbol sequence in the X polarization direction or the Y polarization direction has a good autocorrelation characteristic, and pilot symbol sequences in the two polarization directions have a good cross-correlation characteristic. In addition, an appropriate target polynomial and seed are selected, so that the Q pilot symbols achieve the direct current balance. This helps the receiving end restore the signal quality.

[1028] It should be noted that the data frame considered in embodiments of the present disclosure includes only the pilot symbol and the payload symbol (also referred to as the symbol before framing). This DSP frame structure is different from another existing DSP superframe structure. A DSP superframe usually includes a plurality of types of symbols, for example, a frame synchronization symbol used for frame synchronization and

[illegible]

$x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x350\ 0x01F\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 99\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x130\ 0x014\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 100\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x130\ 0x282\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 101\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x298\ 0x014\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 102\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x298\ 0x282\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 103\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x014\ 0x34C\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 104\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x014\ 0x261\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 105\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x34C\ 0x282\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 106\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x282\ 0x261\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1$

[1033] It should be noted that, when the target polynomial is designed, the degree of the polynomial needs to be constrained not to be greater than 10, and the quantity of non-zero terms of the polynomial needs to be constrained not to be greater than 8. When the polynomial is selected, the degree and the quantity of non-zero terms of the target polynomial need to be selected to be as small as possible, so that the complexity of the generation structure of the pilot symbol is low, and power consumption is low. It is considered that when a target polynomial is given, there may be no seed that can cause a pilot symbol sequence, determined based on the target polynomial and the seed, to have good autocorrelation and cross-correlation. Therefore, in the target polynomial selection, not only a minimum degree and a minimum quantity of non-zero terms are selected, but also whether a corresponding seed can be selected to cause a generated pilot symbol sequence and a real part sequence and an imaginary part sequence of the pilot symbol sequence to have good autocorrelation and cross-correlation characteristics needs to be considered, to help the receiving end restore the signal quality.

[1034] In another possible implementation, the target polynomial is one in the following Table 3.

TABLE-US-00003 TABLE 3 Sequence number Target polynomial 1 $x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 2\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$

[1035] It should be understood that, it is considered that there is a frequency difference between a laser at the transmitting end and a laser at the receiving end, namely, a frequency offset of the laser, and an effect of frame synchronization performed by the receiving end is affected. The polynomials in Table 3 have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is the frequency offset of the laser.

[1036] In a possible implementation, an example in which N=6144, M=64, and Q=96 is used, and a correspondence between the target polynomial, a seed in the X polarization direction, and a seed in the Y polarization direction is one in the following Table 4.

TABLE-US-00004 TABLE 4 Seed in Seed in Se- the X the Y quence polar- polar- num- ization ization ber Target polynomial direction direction 1 $x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}5 + 1\ 0x1D4\ 0x1E1\ 2\ x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}4 + x\{\text{circumflex over ()}\}3 + x + 1\ 0x094\ 0x02B\ 3\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}6 + 1\ 0x058\ 0x3CC\ 4\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}6 + 1\ 0x3CC\ 0x316\ 5\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}6 + 1\ 0x3CC\ 0x18E\ 6\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}6 + 1\ 0x3CC\ 0x163\ 7\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}5 + 1\ 0x152\ 0x14D\ 8\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}5 + 1\ 0x2A4\ 0x14D\ 9\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x + 1\ 0x0B4\ 0x2D1\ 10\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x + 1\ 0x1A2\ 0x05A\ 11\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x + 1\ 0x05A\ 0x2D1\ 12\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x128\ 0x12A\ 13\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x250\ 0x12A\ 14\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x250\ 0x295\ 15\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}4 + x\{\text{circumflex over ()}\}3 + x + 1\ 0x39C\ 0x36E\ 16\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}4 + x\{\text{circumflex over ()}\}3 + x + 1\ 0x18E\ 0x36E\ 17\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}4 + x\{\text{circumflex over ()}\}3 + x + 1\ 0x36E\ 0x239\ 18\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}4 + x\{\text{circumflex over ()}\}3 + x + 1\ 0x36E\ 0x0E7\ 19\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}5 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x232\ 0x36A\ 20\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}5 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x36A\ 0x119\ 21\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x282\ 0x01F\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 22\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + 0x014\ 0x01F\ x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + 1\ 23\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x1BA\ 0x2CE\ 24\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x2A6\ 0x2CE\ 25\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x196\ 0x2CE\ 26\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x2CE\ 0x259\ 27\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x380\ 0x2CE\ 28\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x380\ 0x081\ 29\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x2CE\ 0x375\ 30\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x2CE\ 0x353\ 31\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x2CE\ 0x2FF\ 32\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x2CE\ 0x3FF\ 33\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x301\ 0x2D9\ 34\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x301\ 0x3D9\ 35\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x1BA\ 0x3D9\ 36\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x1BA\ 0x0AD\ 37\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1\ 0x19C\ 0x301\ 38\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x0D4\ 0x0ED\ 39\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x01A\ 0x0ED\ 40\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x3C4\ 0x186\ 41\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x3C4\ 0x261\ 42\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x3C4\ 0x1A3\ 43\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1\ 0x3C4\ 0x0C7\ 44\ x\{\text{circumflex over ()}\}10 +$

$x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x3C4\ 0x346\ 45\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x3C4\ 0x063\ 47\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x3C4\ 0x327\ 48\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x240\ 0x3C4\ 49\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x3C4\ 0x103\ 50\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x2E2\ 0x226\ 51\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x2E2\ 0x165\ 52\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x165\ 0x0C7\ 53\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x266\ 0x0C7\ 54\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x2E2\ 0x3E2\ 55\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}6 + x\{\text{circumflex over ()}\}3 + x\{\text{circumflex over ()}\}2 + x + 1$
 $0x0C4\ 0x226\ 56\ x\{\text{circumflex over ()}\}10 + x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}8 + x\{\text{circumflex over ()}\}7 + x\{\text{circumflex over ()}\}4 + x + 1$
 $0x1D8\ 0x2D9$

[1037] It should be understood that, it is considered that there is the frequency difference between the laser at the transmitting end and the laser at the receiving end, namely, the frequency offset of the laser, and the effect of the frame synchronization performed by the receiving end is affected. The polynomials and the seeds in Table 4 have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is the frequency offset of the laser.

[1038] Further, in a polarization direction, a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in the data frame is less than or equal to 2. In addition, the quantity of pilot symbols that are $-A-A_j$ is the same as the quantity of pilot symbols that are $A+A_j$, and the quantity of pilot symbols that are $-A+A_j$ is the same as the quantity of pilot symbols that are $A-A_j$. This effectively ensures that symbol quantities in each polarization direction are approximately balanced, and can further ensure that a sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality. For example, $Q=96$, the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 23, 25, 25, and 23 in sequence, and the difference between the every two of quantities is less than or equal to 2.

[1039] In a possible implementation, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$; the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$; the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor + 1$; or the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor + 1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$. $\lfloor a \rfloor$ represents rounding down a positive real number a to the nearest integer. For example, $Q=96$, and the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 24, 24, 24, and 24 in sequence, the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 23, 25, 25, and 23 in sequence, or the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 25, 23, 23, and 25 in sequence.

[1040] Further, in two polarization directions, a total quantity of pilot symbols that are $-A-A_j$ is $Q/2$, a total quantity of pilot symbols that are $-A+A_j$ is $Q/2$, a total quantity of pilot symbols that are $A-A_j$ is $Q/2$, and a total quantity of pilot symbols that are $A+A_j$ is $Q/2$. This effectively ensures balance between symbol quantities. In addition, it can be further ensured that the sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[1041] In a possible implementation, in the X polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$. In the Y polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$.

[1042] In another possible implementation, in the X polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor + 1$. In the Y polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor + 1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$.

[1043] Further, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4. A pilot symbol sequence obtained under this condition helps improve the signal quality restored by the receiving end.

[1044] The following describes several specific embodiments by using an example in which $N=6144$, $M=64$, and $Q=96$.

Embodiment 1: Implementation in which the Degree of the Polynomial is 9

[1045] FIG. 9 is a diagram of a structure of another data frame according to an embodiment of this disclosure. Considering DP-16QAM modulation, $96 \times 63 = 6048$ DP payload symbols are obtained through Bose-Chaudhuri-Hocquenghem (BCH) (126, 110) encoding and the DP-16QAM DP modulation. One pilot symbol is inserted before every 63 symbols in the 6048 payload symbols, and a total of 96 pilot symbols are inserted, to obtain the structure of the data frame (also referred to as a DSP frame) shown in FIG. 9, that is, $N=6144$, $M=64$, and $Q=96$. In the X polarization direction or the Y polarization direction, the 96 pilot symbols are generated based on the target polynomial and the seed. For a scenario in which the 96 pilot symbols need to be generated, a consecutive bit sequence $b_{\text{sub}.0}$, $b_{\text{sub}.1}$, $b_{\text{sub}.2}$, ..., and $b_{\text{sub}.191}$ whose bit length is $96 \times 2 = 192$ is obtained based on the target polynomial and the seed. Every two consecutive bits in the bit sequence $b_{\text{sub}.0}$, $b_{\text{sub}.1}$, $b_{\text{sub}.2}$, ..., and $b_{\text{sub}.191}$ are mapped to one symbol, where $b_{\text{sub}.2t}$ and $b_{\text{sub}.2t+1}$ are mapped to one symbol $(2b_{\text{sub}.2t-1}A + (2b_{\text{sub}.2t+1}-1)A_j)$. It should be noted that, alternatively, the symbol $(2b_{\text{sub}.2t-1}A + (2b_{\text{sub}.2t+1}-1)A_j)$ may not be a symbol in a constellation diagram of a used modulation format, and may be one of four symbols in an intermediate area between four outermost symbols and four innermost symbols in the constellation diagram of the used modulation format.

[1046] FIG. 10 is a diagram of a third-generation structure of a pilot symbol according to an embodiment of this disclosure. If the target polynomial is $x\{\text{circumflex over ()}\}9 + x\{\text{circumflex over ()}\}4 + x\{\text{circumflex over ()}\}3 + x + 1$, the seed in the X polarization direction is $0x049$, and the seed in the Y polarization direction is $0x115$, that is, an item whose sequence number is 3 in Table 2 is used, a process of generating the 96 pilot symbols in each of the two polarization directions is shown in FIG. 10.

[1047] In the X polarization direction, an input seed is $0x049$, and is converted into a binary sequence 001001001, that is, values of $m_{\text{sub}.8}$ to $m_{\text{sub}.0}$. If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the X polarization direction is $A-A_j$. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the X polarization direction is $-A-A_j$. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the X polarization direction is $A+A_j$. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the X polarization direction is $-A+A_j$. By analogy, the 96 pilot symbols in the X polarization direction may be obtained.

[1048] In the Y polarization direction, the input polarization seed is $0x115$, and is converted into a binary sequence 100010101, that is, values of

example, differential correlation values of the PS in the X polarization and the Y polarization are separately calculated, so that the polarization directions can be distinguished, frame synchronization and alignment can be performed, and carrier phase recovery is performed by using the PS. The designed pilot symbol sequence has good differential autocorrelation and differential cross-correlation characteristics, and are direct current balanced. This helps the receiving end improve quality of the restored signals.

[1097] The following describes data transmission apparatuses provided in embodiments of this disclosure.

[1098] FIG. 31 is a diagram of a structure of a data transmission apparatus used at a transmitting end according to an embodiment of this disclosure. As shown in FIG. 31, the data transmission apparatus includes a processing unit 101 and a sending unit 102. The processing unit 101 is configured to perform the operation in step 401 in the embodiment shown in FIG. 4. The sending unit 102 is configured to perform the operation in step 402 in the embodiment shown in FIG. 4. For specific operations, refer to related descriptions of the embodiment shown in FIG. 4. Details are not described herein again.

[1099] FIG. 32 is a diagram of a structure of a data transmission apparatus used at a receiving end according to an embodiment of this disclosure. As shown in FIG. 32, the data transmission apparatus includes a processing unit 201 and a receiving unit 202. The receiving unit 202 is configured to receive data frames transmitted from a transmitting end through a channel. The processing unit 201 is configured to perform dispersion compensation, synchronization, phase recovery, and other operations.

[1100] It should be understood that the apparatus provided in this disclosure may alternatively be implemented in another manner. For example, division into the units in the foregoing apparatus is merely logical function division and may be other division in actual implementation. For example, a plurality of units or components may be combined or integrated into another system. In addition, functional units in embodiments of this disclosure may be integrated into one processing unit, may be independent physical units, or two or more functional units may be integrated into one processing unit. The integrated unit may be implemented in a form of hardware, or may be implemented in a form of a software functional unit.

[1101] FIG. 33 is a diagram of another structure of a data transmission apparatus according to an embodiment of this disclosure. As shown in FIG. 33, the data transmission apparatus includes a processor 301, a memory 302, and a transceiver 303. The processor 301, the memory 302, and the transceiver 303 are interconnected through lines. The memory 302 is configured to store program instructions and data. The transceiver 303 is configured to perform data sending and receiving operations, and the processor 301 is configured to perform other operations than data sending and receiving. In a possible implementation, the processor 301 may include the processing unit 101 shown in FIG. 31, and the transceiver 303 includes the sending unit 102 shown in FIG. 31. In another possible implementation, the processor 301 may include the processing unit 201 shown in FIG. 32, and the transceiver 303 includes the receiving unit 202 shown in FIG. 32.

[1102] It should be noted that the processor shown in FIG. 33 may be a general-purpose central processing unit (CPU), a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or another programmable logic device, a transistor logic device, a hardware component, or any combination thereof. The memory shown in FIG. 33 may store an operating system and another application program. When the technical solutions provided in embodiments of this disclosure are implemented by using software or firmware, program code used for implementing the technical solutions provided in embodiments of this disclosure is stored in the memory and is executed by the processor. In an embodiment, the processor may include the memory inside. In another embodiment, the processor and the memory are two independent structures.

[1103] It may be clearly understood by a person skilled in the art that, for the purpose of convenient and brief description, for detailed working processes of the foregoing described system, apparatus, and unit, refer to corresponding processes in the foregoing method embodiments. Details are not described herein again.

[1104] A person of ordinary skill in the art may understand that all or some of the steps in the foregoing embodiments may be implemented by hardware or a program instructing related hardware. The program may be stored in a computer-readable storage medium. The storage medium mentioned above may be a read-only memory, a random-access memory, or the like. Whether the functions are performed by hardware or software depends on particular applications and design constraint conditions of the technical solutions. A person skilled in the art may use different methods to implement the described functions for each particular application, but it should not be considered that the implementation goes beyond the scope of this disclosure.

[1105] When software is used to implement the functions, all or some of the method steps described in the foregoing embodiments may be implemented in a form of a computer program product. The computer program product includes one or more computer instructions. When the computer program instructions are loaded and executed on a computer, the procedures or functions according to embodiments of this disclosure are all or partially generated. The computer may be a general-purpose computer, a dedicated computer, a computer network, or another programmable apparatus. The computer instructions may be stored in a computer-readable storage medium or may be transmitted from a computer-readable storage medium to another computer-readable storage medium. For example, the computer instructions may be transmitted from a website, computer, server, or data center to another website, computer, server, or data center in a wired (for example, a coaxial cable, an optical fiber, or a digital subscriber line (DSL)) or wireless (for example, infrared, radio, or microwave) manner. The computer-readable storage medium may be any usable medium accessible by the computer, or a data storage device, for example, a server or a data center, integrating one or more usable media. The usable medium may be a magnetic medium (for example, a floppy disk, a hard disk, or a magnetic tape), an optical medium (for example, a digital versatile disc (DVD)), a semiconductor medium (for example, a solid-state drive (SSD)), or the like.

Claims

1. A data transmission method, comprising: generating data frames, wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and M-1 payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of four complex numbers: $-A-Aj$, $-A+Aj$, $A-Aj$, or $A+Aj$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and sending the data frames.
2. The method of claim 1, wherein the target polynomial is $x^{\circ}9+x^{\circ}8+x^{\circ}5+x^{\circ}4+1$.
3. The method of claim 1, wherein each of the Q pilot symbols is at a start location of one of the M consecutive symbols in which the pilot symbol is located.
4. The method of claim 1, wherein a first sequence comprising Q pilot symbols in a first polarization direction is different from a second sequence comprising Q pilot symbols in a second polarization direction, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.
5. The method of claim 1, wherein a correspondence among the target polynomial, a first seed in a first polarization direction, and a second seed in a second polarization direction is: the target polynomial is $x^{\circ}9+x^{\circ}8+x^{\circ}5+x^{\circ}4+1$; the first seed is 0x175; and the second seed is 0x03D.
6. The method of claim 5, wherein $Q=96$.
7. The method of claim 6, wherein $N=6144$, and wherein $M=64$.
8. The method of claim 1, wherein in each polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or

wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and $M-1$ payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of four complex numbers: $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and a transmitter coupled to the first processor and configured to send the data frames; and a second data transmission apparatus connected to the first data transmission apparatus and comprising: a receiver configured to receive the data frames; and a second processor coupled to the receiver and configured to process the data frames.
